MILITARY SPECIFICATION

TEST SET SUBASSEMBLY: 10554898
(Pulse Generator)

1. SCOPE

This specification covers a test set subassembly, hereinafter called pulse generator, used in Test Set Subassembly MX-8050/TPM-22, part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Process and Inspection of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10554898	Pulse Generator

PACKAGING DATA SHEET

10554898	Packaging of Pulse Generator

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The pulse generator shall be manufactured in accordance with Drawing F10554898 and drawings pertaining thereto.

3.2 General specification. - Where applicable, the pulse generator shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

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3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The pulse generator shall be mounted on the vibration table in normal service position. The pulse generator shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The pulse generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. - The pulse generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

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3.3.3.2 Nonoperating. - The pulse generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The pulse generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage). - The pulse generator shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-nonoperating. - The pulse generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance. - When applied with the inputs specified in Table I, the pulse generator shall yield the outputs specified in Table I.

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TABLE I - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTIC	TERMINALS
Input voltage		
+5V	+5.0 ± 0.5 Vdc	P1-30, 31, 32, (+); P1-24, 25, 26 (RET)
Input signals		
7.5 mc	7.5 mc square wave or sine wave, amplitude 2.4-5.2V peak-to-peak.	P1-43
$\overline{\text{STROBE}}$	Approx 940 kc negative pulse, amplitude from 3.9 ± 1.1V level to OV and return, pulse width approx 133 ns.	P1-1, 9, 10, 12, 16
$\overline{\text{T1}}$	20 kc negative pulse, amplitude from 3.9 ± 1.1V level to OV and return, pulse width approx 0.8 usec.	P1-22
$\overline{\text{T3}}$	Negative pulse, same frequency, and width as $\overline{\text{T1}}$, delayed approx 2.14 usec from leading edge of $\overline{\text{T1}}$.	P1-15

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TABLE I - INPUT AND OUTPUT CHARACTERISTICS (Cont'd)

REQUIREMENT	CHARACTERISTIC	TERMINALS
Output signals		
Cp	7.5 mc square wave, delayed approx 10 ns from 7.5 mc input signal, into 150 ohms.	P1-47
Cpd	Same as Cp, except delayed 20-40 ns from Cp, into 150 ohms.	P1-45
$\overline{\text{STROBE}}$	Same as $\overline{\text{STROBE}}$ input signal except delayed 20-40 ns, into 150 ohms.	P1-3, 4, 54, 44, and 2, respectively, to inputs at P1-1, 9, 10, 12, and 16.
STROBE	Inverse of $\overline{\text{STROBE}}$ input signal, except delayed 10-20 ns, into 150 ohms.	P1-53, 41, and 46, respectively, to inputs at P1-10, 12, and 16.
$\overline{\text{T1}}$	Same as $\overline{\text{T1}}$ input signal, except delayed 20-40 ns, into 150 ohms.	P1-34
T1	Inverse of $\overline{\text{T1}}$ input signal, except delayed 10-20 ns, into 150 ohms.	P1-40
$\overline{\text{T3}}$	Same as $\overline{\text{T3}}$ input signal, except delayed 20-40 ns, into 150 ohms.	P1-19

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TABLE I - INPUT AND OUTPUT CHARACTERISTICS (Cont'd)

REQUIREMENT	CHARACTERISTIC	TERMINALS
T3	Inverse of $\overline{T3}$ input signal, except delayed 10-20 ns, into 150 ohms.	P1-18
$\overline{T\ 1/4\ KHz}$	Negative pulse, exactly 1/5 of frequency of $\overline{T1}$ input signal; amplitude from 3.9 ± 1.1V level to OV and return; pulse width approx 0.8 usec; approx coincident with every fifth $\overline{T1}$ pulse, into 150 ohms.	P1-33
$\overline{F\ 4/5}$	Negative pulse, exactly 1/5 of frequency of $\overline{T1}$ input signal; amplitude from 3.9 ± 1.1V level to OV and return; pulse width approx 2.14 usec, leading edge approx coincident with trailing edge of $\overline{T1}$ input signal.	P1-35
$\overline{F\ 1/5}$	Complex waveform per Fig. 1	P1-39
F 1/5	Inverse of $\overline{F\ 1/5}$	P1-38
$\overline{F\ 2/5}$	Complex waveform per Fig. 1	P1-37
F 2/5	Inverse of $\overline{F\ 2/5}$	P1-36

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one pulse generator manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One pulse generator out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

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TABLE III - SPECIAL SAMPLING

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specifications	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment. - In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

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TEST EQUIPMENT LIST

ITEM	DESCRIPTION
1. Power Supply	5.0 ± 0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 300 mA.
2. Signal Generator	7.5 mc square wave or sine wave, amplitude 5.0 ± 0.2 V peak-to-peak, into 50 ohm load.
3. Pulse Generator	0.9-1.0 mc repetition rate. Negative-going pulses from +3 Vdc level to OV and return into 150 ohms. Pulse width approx 133 ns. Must also be capable of producing inverse of the above.
4. Pulse Generator	19-20 kc repetition rate. Negative-going pulses from +3 Vdc level to OV and return, into 150 ohms. Pulse width 0.2-0.8 usec. Must have trigger output capability compatible with external drive requirements of item 5. Trigger out must be coincident with output pulse.
5. Pulse Generator	19-20 kc repetition rate. Negative-going pulses from +3 Vdc level to O V and return. Pulse width approx 0.8 usec. Must have external drive input compatible with trigger output capability of item 4. Output must be capable of being delayed 2.0-2.3 usec from external drive input.

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TEST EQUIPMENT LIST - continued

ITEM	DESCRIPTION
6. Oscilloscope	Dual channel, vertical pass-band DC to 15 MHz. Rise time of scope plus measuring probe 10 nanoseconds or less. Sweep period 50 nanoseconds per centimeter or less. Input capacitance less than 25 picofarads.
7. Test harness	Per Figure 2.

PARTS LIST FOR TEST HARNESS, ITEM 7

DESIGNATION	PART	DESCRIPTION
J1	Connector	Malco 5893232-1, Elco 07-5422-0198-01, or equivalent
R1	Resistor	150 ohms, ±10%, 1/2 W

Terminal strips, wire, per Figure 2.

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of Method 514 of MIL-STD-810 do not apply. After vibration, the pulse generator shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

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4.6.1.2 Shock. - This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the pulse generator shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating. - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the pulse generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the pulse generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating. - This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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4.6.3 Performance. - Adjust power supply, item 1; signal generator, item 2; and pulse generators, items 3, 4, and 5; to deliver input signals indicated on Figure 2 and specified in Table I, and connect to test harness, item 7. Turn off test equipment. Insert pulse generator in connector J1 of test harness. Turn on test equipment. Use channel B of oscilloscope, item 6, to monitor output signal, and channel A to monitor reference signal, in the following procedure.

Connect lead Y (Fig. 2) to test harness terminal TB1-43, and monitor 7.5 mc input signal on channel A of oscilloscope, item 6. Connect lead Z (Fig. 2) to test harness terminal TB2-47, and verify that Cp output signal meets requirements specified in Table I. Connect lead Z to test harness terminal TB2-45, and verify that Cpd output signal meets requirements specified in Table I.

With leads X, Y, and Z connected to test harness terminals shown below, verify that STROBE and $\overline{\text{STROBE}}$ output signals meet requirements specified in Table I.

Lead X and Y on	Lead Z on	Output Signal
TB1-1	TB2-3	$\overline{\text{STROBE}}$
TB1-9	TB2-4	"
TB1-10	TB2-54	"
TB1-12	TB2-44	"
TB1-16	TB2-2	"
TB1-10	TB2-53	STROBE
TB1-12	TB2-41	"
TB1-16	TB2-46	"

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With leads Y and Z connected to the test harness terminals shown below, verify that the indicated output signals meet requirements specified in Table I.

Lead Y on	Lead Z on	Output Signal
TB1-22	TB2-34	$\overline{T1}$
TB1-22	TB2-40	T1
TB1-15	TB2-19	$\overline{T3}$
TB1-15	TB2-18	T3
TB1-22	TB2-33	$\overline{T\ 1/4\ KHz}$
TB1-22	TB2-35	$\overline{F\ 4/5}$
TB1-22	TB2-39	$\overline{F\ 1/5}$
TB1-22	TB2-38	F 1/5
TB1-22	TB2-37	$\overline{F\ 2/5}$
TB1-22	TB2-36	F 2/5

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10554898. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The pulse generator provides buffering and generates the inverses of its input signals. It also uses two of its input signals to generate several complex waveforms. All of its output signals are subsequently fed to line drivers and are then used as test signals in Test Set, Subassembly MX-8050/TPM-22. The latter subassembly is used to test printed circuit cards.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging and packing.

(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each pulse generator.

Custodian

Army - MU

Preparing Activity

Army - MU

Project No. 4931-A032

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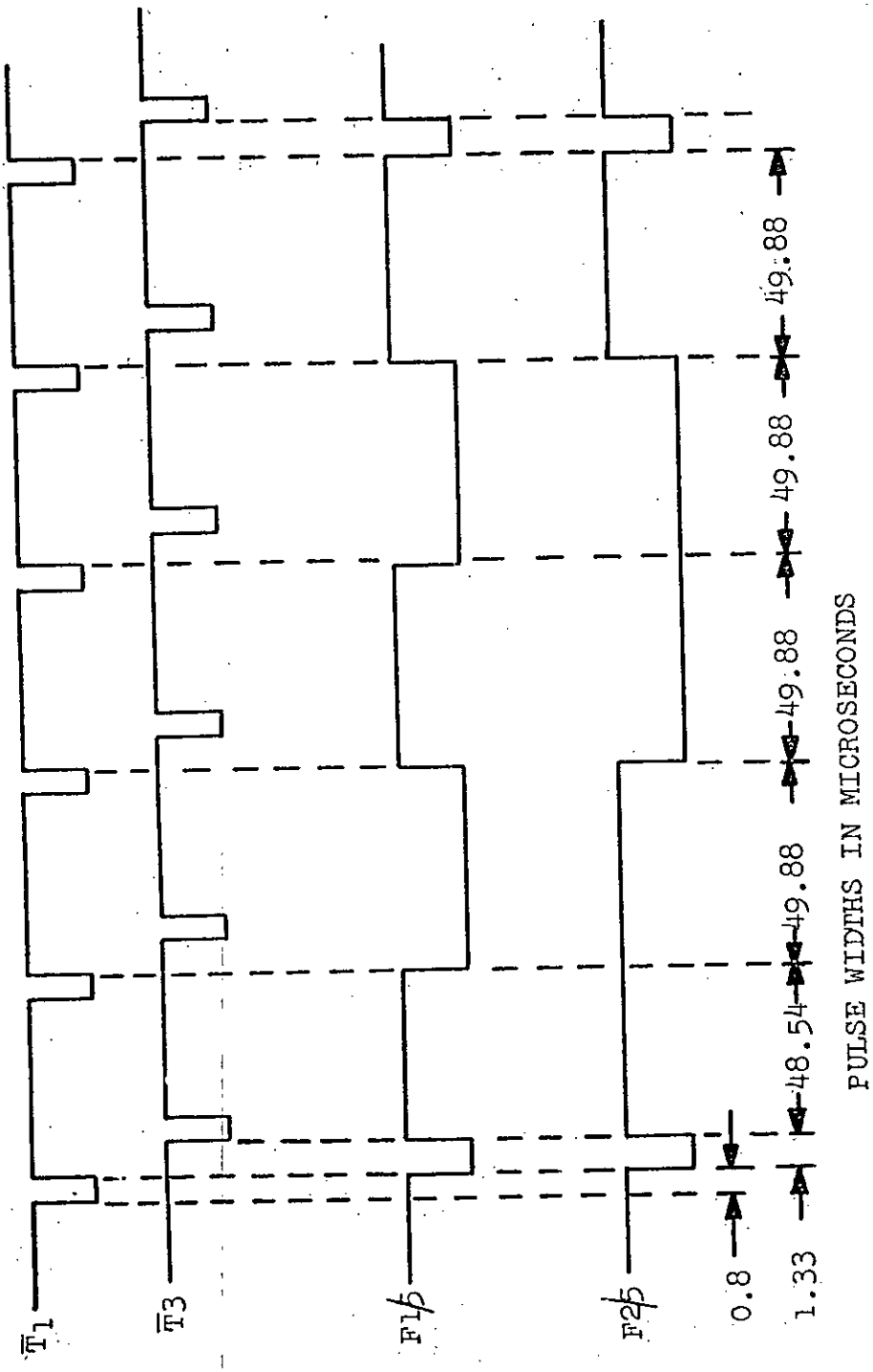


FIGURE 1 F 1/5 AND F 2/5 WAVEFORMS

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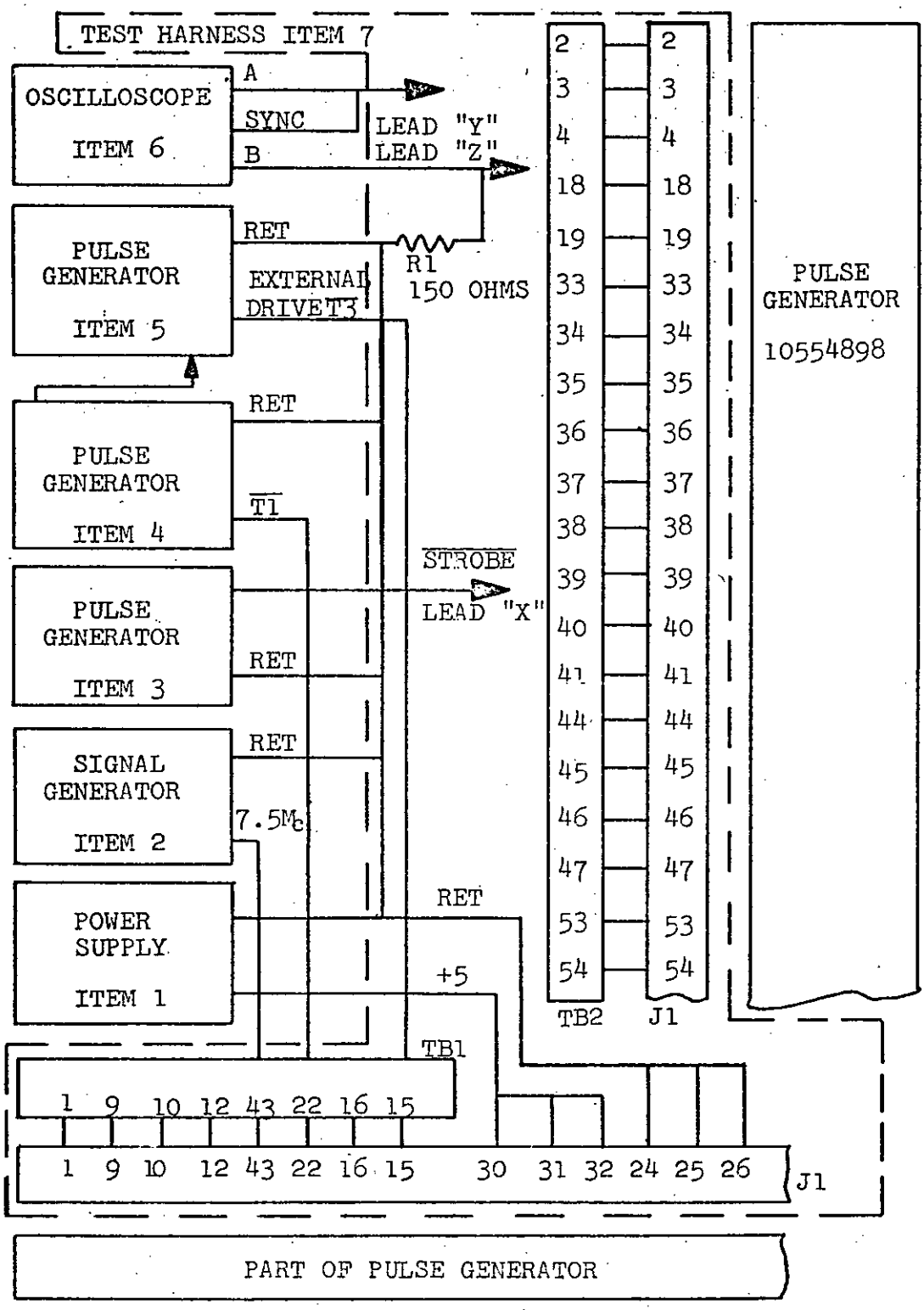


FIGURE 2 TEST SETUP FOR PULSE GENERATOR

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Frankford Arsenal
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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